

# Distribution-Shifting: Novel Phase-distortion Metrics for Hologram Processing

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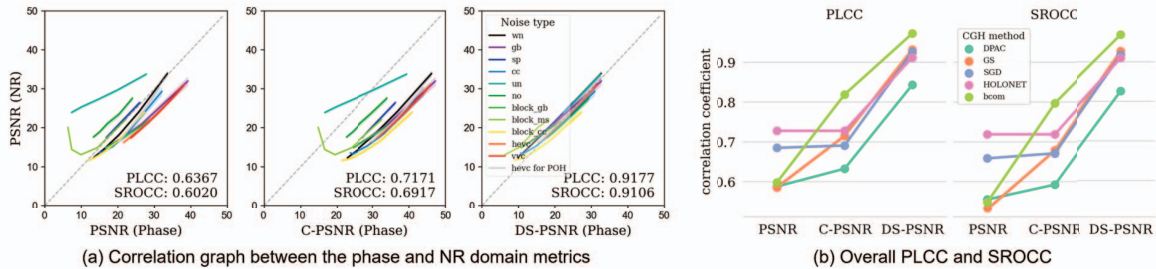


Figure 1: (a) and (b) show the average correlation performance between the peak signal-to-noise ratio (PSNR) in numerical reconstruction (NR) domain and the PSNR in phase domain across four phase-only hologram (PoH) datasets (DPAC, GS, SGD, HoloNet) and a 3D dataset (B-com). The results indicate that our proposed PSNR significantly improve both the pearson linear correlation coefficient (PLCC) and spearman rank order correlation coefficient (SROCC) performances for all datasets.

## ABSTRACT

Due to the lack of suitable phase distortion metrics, the optimization and evaluation of phase-holograms have relied on numerical reconstruction (NR) domain metrics. However, the necessity of NR for hologram processing leads to more intricate design and a higher computational burden. In this paper, we introduce a distribution-shifting (DS) algorithm to enable optimizing or measuring distortions directly in phase-domain, considering  $2\pi$ -periodicity and shift-invariance. Experimental results with various noise types demonstrate a strong correlation between phase-domain metric with DS and their NR-domain counterparts. We believe that our DS-metric could facilitate direct optimization approaches in various phase-hologram processing techniques.

**Index Terms:** Computing methodologies—Computer graphics—Image manipulation—Image processing;

## 1 INTRODUCTION

Compared with amplitude-only holograms (AoHs), phase-only holograms (PoHs) are more frequently utilized due to their higher diffraction efficiency. With the importance of the PoHs, many investigations have been conducted to generate high-quality PoHs from full-complex holograms such as DPAC [4], GS [3], SGD [6], and HoloNet [6]. In general, including the aforementioned methods, the optimization or evaluation of PoHs is typically performed in the numerical reconstruction (NR) domain. This is driven by the

importance of reconstruction quality over the hologram domain and the lack of suitable phase distortion metrics. However, the computational complexity of the NR process is very high, in general. Furthermore, NR is not applicable to certain applications such as when compressing a phase image using block-based standard codecs like VVC. Note that simply applying conventional image quality assessment (IQA) metrics, such as mean square error (MSE) and peak signal-to-noise ratio (PSNR), to phase holograms could not be a solution due to their poor correlation with the metric scores in the NR domain.

In this paper, we propose a novel phase distortion metric named **Distribution-Shifted metric (DS-metric)**. This metric is designed to address the  $2\pi$ -periodicity and shift-invariance property in phase signals. Experimental results on diverse Phase-hologram datasets with various noise types are summarized in Fig. 1, where a strong correlation is demonstrated between the proposed phase-domain PSNR and their NR-domain counterpart.

## 2 EVALUATION PROTOCOL AND DATABASE

We aim to propose a reliable phase-domain metric that strongly correlates with the NR-domain metric. To evaluate the correlation, we use the pearson linear correlation coefficient (PLCC) [2] and the spearman rank order correlation coefficient (SROCC) [7]. To assess the correlation performances of our phase distortion metrics, we constructed a dedicated phase hologram database named Noisy Phase Image Database (NPI-DB). The definition and detailed descriptions of NPI-DB can be found in the [1].

## 3 PROPOSED PHASE-DISTORTION METRICS

### 3.1 Design requirements

**$2\pi$ -periodicity** The authors in [5] introduced a new phase difference measure that incorporates the  $2\pi$ -periodicity of the phase signal as referred to the shorter circular difference (SCD). The SCD image  $\tilde{D}$  is calculated by adjusting the difference values to be smaller

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than  $\pi$  at all position as shown in Eq. (1).

$$\tilde{D}(m, n) = (X(m, n) - Y(m, n) + \pi) \% 2\pi - \pi, \quad (1)$$

where  $X$  and  $Y$  represent the reference and noisy hologram images, respectively. The variables  $m$  and  $n$  represent the spatial position. Following the SCD concept, the circular PSNR (C-PSNR) proposed by [5], is defined in Eq.(3).

$$C-MSE(X, Y) = E[\tilde{D}^2], \quad (2)$$

$$C-PSNR = 10 \cdot \log_{10} \left( \frac{MAX_I^2}{C-MSE(X, Y)} \right), \quad (3)$$

**Shift-invariance** There can be an infinite number of phase holograms that yield the same NR results. This is due to the ‘**shift-invariance**’ property of the hologram, which means that global phase shifting by the same value  $\Delta$  regardless of its spatial position has no effect on NR result, as can be formulated by Eq. (4). Unfortunately, this property results in significantly poor performance when measured using conventional image distortion metrics in the phase domain. Therefore, the phase distortion metrics should be designed to be invariant under such global phase shifting.

$$NR(X(m, n)) \equiv NR((X(m, n) + \Delta) \% 2\pi). \quad (4)$$

### 3.2 Distribution-shifted (DS) metric

To ensure shift-invariance, we first align the centers of  $X$  and  $Y$  using the proposed distribution-shifting process. Then we separate each of the aligned phase signals  $\hat{X}$  and  $\hat{Y}$  into real and imaginary components  $\hat{X}_{re}$ ,  $\hat{X}_{im}$ ,  $\hat{Y}_{re}$ ,  $\hat{Y}_{im}$  and take the root mean square (RMS) of the two metric scores, each calculated for the real and imaginary parts. Since the real and imaginary part of a phase signal does not follow the  $2\pi$ -periodicity, we can easily convert any metric into DS-metric in this way. The overall pipeline for the proposed DS-metric computation process is summarized in Fig.2.

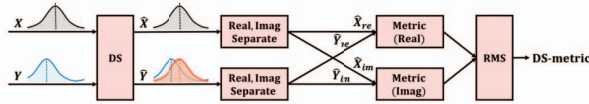


Figure 2: The DS-metric pipeline.

**Distribution-shifting** Our approach focuses to accurately align the distribution centers of two phase images by compensating for the offset between the noisy phase image  $Y$  and the reference phase image  $X$  before measuring distortion. To determine the compensation offset  $E[\tilde{D}]$ , we calculate the SCD  $\tilde{D}$  between the  $X$  and  $Y$  using Eq.(1) and then shift the  $Y$  by the  $E[\tilde{D}]$  for the entire position as illustrated in Fig.3-(a). If the shifted center aligns with the opposite semicircle, an additional  $\pi$ -shifting is applied as shown in Fig.3-(b). The  $E[|\tilde{D}|]$  can provides insights into which semicircle the distribution is aligned. If  $|E[|\tilde{D}|] - |E[\tilde{D}]||$  is greater than  $\pi/2$ , an additional  $\pi$ -shifting is performed, as formulated by Eq.(5).

$$\tilde{Y}(m, n) = \begin{cases} (Y(m, n) - E[\tilde{D}]) \% 2\pi, & |E[|\tilde{D}|] - |E[\tilde{D}]|| < \frac{\pi}{2}, \\ (Y(m, n) - E[\tilde{D}] + \pi) \% 2\pi, & \text{otherwise.} \end{cases} \quad (5)$$

Although the distributions have been effectively shifted to align their centers, the centers have not been explicitly aligned with the zero-axis. To make the distribution of the two phase images be zero-centered, mean values from the respective images are subtracted, as shown in Eq.(6).

$$\begin{aligned} \hat{X}(m, n) &= (X(m, n) - E[X]) \% 2\pi \\ \hat{Y}(m, n) &= (\tilde{Y}(m, n) - E[\tilde{Y}]) \% 2\pi \end{aligned} \quad (6)$$

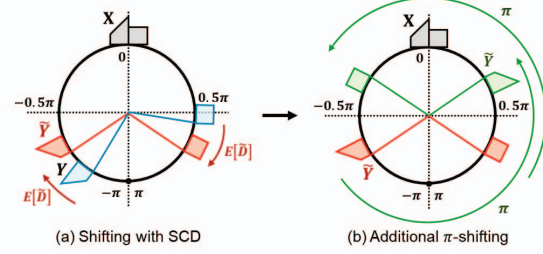


Figure 3: Example of Distribution-shifting.

**DS-metric calculation** After distribution-shifting, we calculate the DS-MSE by calculating the RMS of the MSEs of the real and imaginary parts as given in Eq. (7) and (8).

$$\hat{I}_{re} = \cos(\hat{I}), \quad \hat{I}_{im} = \sin(\hat{I}), \quad \text{where } \hat{I} \in \{\hat{X}, \hat{Y}\} \quad (7)$$

$$DS-MSE(X, Y) = \sqrt{\frac{MSE(\hat{X}_{re}, \hat{Y}_{re})^2 + MSE(\hat{X}_{im}, \hat{Y}_{im})^2}{2}}, \quad (8)$$

$$DS-PSNR(X, Y) = 10 \cdot \log_{10} \left( \frac{MAX_I^2}{DS-MSE(X, Y)} \right). \quad (9)$$

### 4 EXPERIMENTAL RESULTS ON DS-METRICS

Fig.1-(a), shows correlation plots where the x-axis corresponds to phase domain metrics and the y-axis represents NR domain metrics for various different noise type on NPI-DB. A graph closer to linear indicates better performance. PSNR and C-PSNR exhibit varying slopes for different noise types, results in lower overall PLCC and SROCC performance across all data points as shown in Fig.1-(b). On the other hand, DS-PSNR exhibits a notable trend towards linearity with consistent slopes across various noise types.

### 5 CONCLUSIONS

Proposed method outperforms PSNR and C-PSNR across different noise types, allowing the phase domain to be the primary processing domain and reducing computational costs. We believe that our DS-metric could facilitate direct optimization approaches in various phase-hologram processing techniques.

### ACKNOWLEDGMENTS

This work was supported by Samsung Research Funding & Incubation Center of Samsung Electronics under Project Number SRFC-IT2201-03 (97%), and Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (No.RS-2022-00155911, Artificial Intelligence Convergence Innovation Human Resources Development (Kyung Hee University)) (3%).

### REFERENCES

- [1] S. Choi, J. Y. Cha, H. Ban, K.-J. Oh, H. Ko, and H. Y. Kim. Npi-database. <https://github.com/VML-lambda/Hologram-NPI-DB>, 2024.
- [2] D. Freedman, R. Pisani, and R. Purves. Statistics (international student edition). *Pisani, R. Purves, 4th edn. WW Norton & Company, New York*, 2007.
- [3] R. W. Gerchberg. A practical algorithm for the determination of plane from image and diffraction pictures. *Optik*, 35(2):237–246, 1972.
- [4] C.-K. Hsueh and A. A. Sawchuk. Computer-generated double-phase holograms. *Applied optics*, 17(24):3874–3883, 1978.
- [5] K.-J. Oh, H. Ban, S. Choi, H. Ko, and H. Y. Kim. Hevc extension for phase hologram compression. *Optics Express*, 31(6):9146–9164, 2023.
- [6] Y. Peng, S. Choi, N. Padmanaban, and G. Wetzstein. Neural holography with camera-in-the-loop training. *ACM Transactions on Graphics (TOG)*, 39(6):1–14, 2020.
- [7] J. H. Zar. Spearman rank correlation. *Encyclopedia of Biostatistics*, 7, 2005.